

LAB MANUAL
of
INFORMATION SECURITY SYSTEMS LABORATORY
B. TECH (CSE) 6th SEMESTER
SUBJECT CODE - CSDC 0334



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[illegible]



Lab Assignment 1

Question 1. Write the program to find prime numbers between 1 to 1000 and also print count.

Code:

```
#include <iostream>
#include <vector>
using namespace std;
int sieve(int n) {
    vector<bool> isPrime(n + 1, true);
    isPrime[0] = isPrime[1] = false;

    for (int i = 2; i * i <= n; i++) {
        if (isPrime[i]) {
            for (int j = i * i; j <= n; j += i) {
                isPrime[j] = false;
            }
        }
    }

    int count = 0;
    for (int i = 2; i <= n; i++) {
        if (isPrime[i]) {
            cout << i << " ";
            count++;
        }
    }
    cout << endl;
    return count;
}
int main() {
    int n = 1000;
    cout << "Total primes = " << sieve(n) << endl;
}
```

Output:

```
PS C:\Users\mansil\OneDrive\Desktop\c++\sa> cd "c:\Users\mansil\OneDrive\Desktop\c++\sa\" ; if ($?) { g++ cdlab3.cpp -o cdlab3 } ; if ($?) { .\cdlab3 }
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199 211 223 227
229 233 239 241 251 257 263 269 271 277 281 283 293 307 311 313 317 331 337 347 349 353 359 367 373 379 383 389 397 401 409 419 421 431 433 439 443 449 457 461 463
467 479 487 491 499 503 509 521 523 541 547 557 563 569 571 577 587 593 599 601 607 613 617 619 631 641 643 647 653 659 661 673 677 683 691 701 709 719 727 733 739
743 751 757 761 769 773 787 797 809 811 821 823 827 829 839 853 857 859 863 877 881 883 887 907 911 919 929 937 941 947 953 967 971 977 983 991 997
Total primes = 168
```

Question 2. Write the program to implement matrix multiplication of 3*3 matrix.

Code:

```
#include <iostream>
using namespace std;
int main() {
    int m, n, p;
    cin >> m >> n >> p;
    int A[m][n], B[n][p], C[m][p];
```

```

for(int i = 0; i < m; i++)
    for(int j = 0; j < n; j++)
        cin >> A[i][j];
for(int i = 0; i < n; i++)
    for(int j = 0; j < p; j++)
        cin >> B[i][j];
for(int i = 0; i < m; i++)
    for(int j = 0; j < p; j++)
        C[i][j] = 0;
for(int i = 0; i < m; i++) {
    for(int j = 0; j < p; j++) {
        for(int k = 0; k < n; k++) {
            C[i][j] += A[i][k] * B[k][j];
        }
    }
}
for(int i = 0; i < m; i++) {
    for(int j = 0; j < p; j++) {
        cout << C[i][j] << " ";
    }
    cout << endl;
}

```

Output:

```
PS C:\Users\manshi\OneDrive\Desktop\c+\sa> cd "c:\Users\man
```

```

2 2 2
1 2 3 4
1 2 3 4
7 10
15 22

```

Question 3. Write the program to find the inverse of 3*3 Matrix.

Code:

```

#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    double A[n][2*n];
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < n; j++)
            cin >> A[i][j];
    }
}

```

```

for(int j = n; j < 2*n; j++)
    A[i][j] = (i == j-n);}
for(int i = 0; i < n; i++) {
    double diag = A[i][i];
    if(diag == 0) {
        cout << "Inverse does not exist";
        return 0;
    }
    for(int j = 0; j < 2*n; j++)
        A[i][j] /= diag;
    for(int k = 0; k < n; k++) {
        if(k != i) {
            double factor = A[k][i];
            for(int j = 0; j < 2*n; j++)
                A[k][j] -= factor * A[i][j];
        }}
    cout << "Inverse Matrix:\n";
    for(int i = 0; i < n; i++) {
        for(int j = n; j < 2*n; j++)
            cout << A[i][j] << " ";
        cout << endl;
    }
}

```

Output:

```

PS C:\Users\manshi\OneDrive\Desktop\c+\sa> cd "c:\Use
tempCodeRunnerFile }
3
1 2 3 12 2 3 2 3 4
Inverse Matrix:
-0.0909091 0.0909091 0
-3.81818 -0.181818 3
2.90909 0.0909091 -2

```

Lab Assignment 2

Question 1. Write a program to implement a substitution cipher.

SOURCE CODE:

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    string all_letrs = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ";
    map<char, char> dict1; int key = 4;
    for (int i = 0; i < all_letrs.length(); i++) {
        dict1[all_letrs[i]] = all_letrs[(i + key) % all_letrs.length()];
    }
    string plain_txt = "I am studying Data Encryption"; string cipher_txt;

    for (char& c : plain_txt) {
        if (dict1.find(c) != dict1.end()) { cipher_txt += dict1[c];
        }
        else {
            cipher_txt += c;
        }
    }
    cout << "Cipher Text is: " << cipher_txt << endl; map<char, char> dict2;
    for (int i = 0; i < all_letrs.length(); i++) { dict2[all_letrs[i]]
    = all_letrs[(i - key) % all_letrs.length()];
    }
    string decrypt_txt;
    for (char& c : cipher_txt) {
        if (dict2.find(c) != dict2.end()) { decrypt_txt += dict2[c];
        }
        else {
            decrypt_txt += c;
        }
    }
    cout << "Recovered plain text : " << decrypt_txt << endl; return 0;
```

OUTPUT:

```
Cipher Text is: M eq wxyhCmrk Hexe IrgvCtxmsr
Recovered plain text : I am studying Data Encryption
```

Question 2. Write a program to implement a Frequency Analysis of given file.

SOURCE CODE:

```
#include <iostream>
#include <fstream>
#include <cctype>
#include <iomanip>
using namespace std;

int main() { ifstream file; string filename;
int freq[26] = {0}; char ch;
int totalLetters = 0;
```

```
cout << "Enter file name: "; cin >> filename;
```

```
file.open(filename); if (!file) {  
cout << "Error opening file!" << endl;  
return 0;  
}
```

```
while (file.get(ch)) { if (isalpha(ch)) {  
ch = tolower(ch); freq[ch - 'a']++; totalLetters++;  
}  
}
```

```
file.close();  
cout << "\nLetter Frequency Analysis:\n"; cout << "Letter  Count  Percentage\n";
```

```
for (int i = 0; i < 26; i++) {  
double percent = (totalLetters == 0) ? 0 : (freq[i] * 100.0) / totalLetters; cout << char('A' + i) << " "<<  
freq[i] << " "  
<< fixed << setprecision(2) << percent << "%\n";  
}return 0;  
}
```

OUTPUT:

```
Enter file name: issfile.txt  
  
Letter Frequency Analysis:  
Letter  Count  Percentage  
A        0      0.00%  
B        1      1.64%  
C        0      0.00%  
D        3      4.92%  
E        4      6.56%  
F        7     11.48%  
G       15     24.59%  
H        2      3.28%  
I        1      1.64%  
J        0      0.00%  
K        0      0.00%  
L        0      0.00%  
M        0      0.00%  
N        0      0.00%  
O        0      0.00%  
P        0      0.00%  
Q        0      0.00%  
R        2      3.28%  
S        0      0.00%  
T        2      3.28%  
U        5      8.20%  
V        6      9.84%  
W        0      0.00%  
X        0      0.00%  
Y       13     21.31%  
Z        0      0.00%
```

Question 3. Write a program to implement an affine cipher.

SOURCE CODE:

```
#include<bits/stdc++.h> using namespace std;
```

```
const int a = 17; const int b = 20;
```

```
string encryptMessage(string msg)
```

```

{
string cipher = "";
for (int i = 0; i < msg.length(); i++) { if(msg[i]!=' ')
cipher = cipher + (char) (((a * (msg[i]-'A') ) + b) % 26) + 'A'); else
cipher += msg[i];
}
return cipher;}
int main(void)
{
string msg = "AFFINE CIPHER";

string cipherText = encryptMessage(msg);
cout << "Encrypted Message is : " << cipherText<<endl;
return 0;
}

```

OUTPUT:

```

Encrypted Message is : UBBAHK CAPJKX

```